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MP Ops - Assignment 2

Hugging Face Fine-Tuning, Experiment Tracking & Model Deployment

1. Model Selection Rationale

DistilBERT (distilbert-base-cased) was selected due to its strong balance between performance, speed, and computational efficiency. The Goodreads review dataset requires contextual understanding of language, sentiment, and written patterns. These are areas where transformer-based models tend to excel. Being built on BERT's architecture, DistilBERT captures contextual meaning effectively. Therefore, making it well-suited for multi-class text classification.

Compared to full BERT, DistilBERT is approximately 40% smaller and 60% faster while retaining around 97% of its language understanding capability. This makes it ideal for Kaggle's free GPU environment where computational resources are limited, reducing both training time and memory usage significantly.

The cased variant was chosen because book review text frequently contains proper nouns and genre-specific terminology where capitalisation carries semantic meaning. Pre-training on English Wikipedia and BookCorpus further supports its suitability for literary and review-style language.

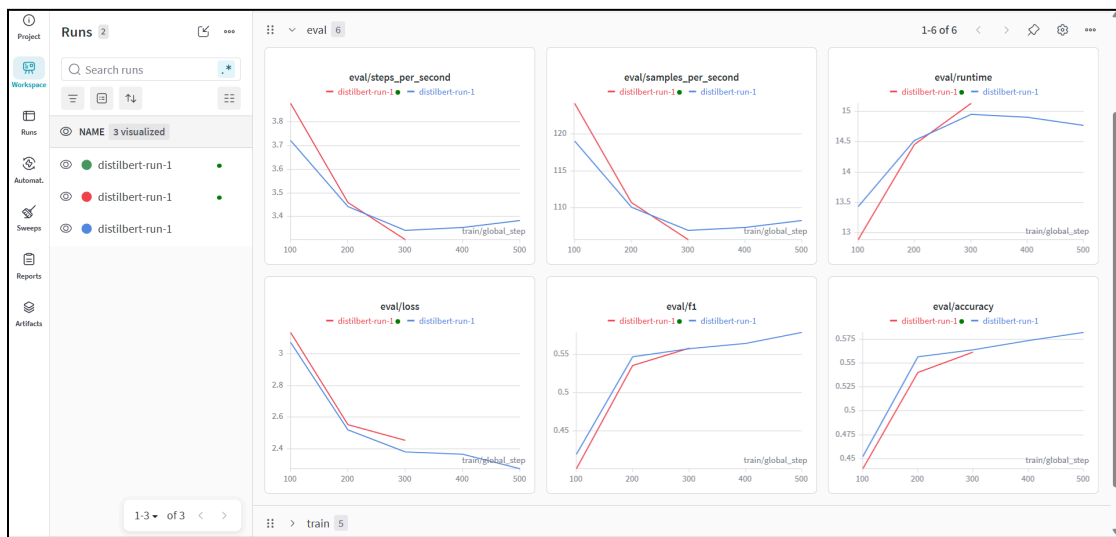
2. Kaggle training setup

- a. Importing the Notebook: The starter notebook was downloaded from the provided Colab link and imported into a new Kaggle Notebook using File → Import Notebook. This allowed the full training workflow to run on Kaggle's cloud infrastructure with free GPU access.
- b. GPU Configuration: GPU acceleration was enabled under Settings → Accelerator → GPU T4 x2. Internet access was also toggled on to allow package installation and pushing to Hugging Face Hub. Kaggle provides up to 30 hours of free GPU per week, which was good for this training run.
- c. Secrets Management: API credentials were stored securely using Kaggle's built-in Secrets manager by going to Add-ons → Secrets, rather than hardcoding them in the notebook. Two secrets were configured:
 - WANDB_API_KEY: retrieved from wandb.ai/settings
 - HF_TOKEN: retrieved from huggingface.co/settings/tokens

These were loaded at runtime using the `kaggle_secrets` library and set as environment variables, keeping credentials secure while enabling seamless integration with both W&B and Hugging Face Hub.

- d. W&B Integration: Weights & Biases was initialised using `wandb.init()` with the project name “mlops-assignment2” and a descriptive run name. The `TrainingArguments` object was configured with `report_to="wandb"`, which enabled automatic logging of training loss, validation loss, accuracy, F1 score, and learning rate at every evaluation step throughout the training run.

W&B training dashboard



3. Challenges:

- a. W&B Integration with the Trainer: The most significant technical challenge encountered was the W&B session lifecycle. The Hugging Face Trainer with `report_to="wandb"` automatically calls `wandb.finish()` at the end of `trainer.train()`, which closes the active run. Any subsequent `wandb.log()` call in a separate cell then throws a "You must call `wandb.init()` before `wandb.log()`" error. This was resolved by re-initialising the W&B run with `resume="allow"` before the evaluation cell, which re-attaches to the same run without creating a duplicate entry on the dashboard.
- b. Kaggle Secrets and Internet Access: Initially, API calls to Hugging Face Hub failed because the Internet toggle in Kaggle's session options was disabled. Enabling Internet access under Settings resolved this. Storing credentials as Kaggle Secrets rather than hardcoding them in the notebook was straightforward once the Add-ons → Secrets interface was located.

4. Learnings

- a. Experiment tracking with W&B makes it easy to compare runs and catch issues like diverging loss early.
- b. Kaggle Secrets allows a clean, secure alternative to environment variables for managing API keys in shared notebooks.
- c. Pushing a trained model to Hugging Face Hub with `push_to_hub()` is remarkably simple and makes the model instantly accessible to anyone.

5. Links

W&B	mlops-assignment2 on W&B
HF	pjayG25AIT2045/distilbert-goodreads-genres
GitHub	MLOPS-Assignment-2
Kaggle	mlops-assignment-2 on Kaggle